



UNIVERSITY MALAYSIA OF COMPUTER SCIENCE & ENGINEERING (UNIMY)

Programme	Bachelor of Computer Science with Honours (BCS)
Course Name	Distributed Application Development
Course Code	BCCS3063
Credit Hours	3
Lecturer Email	trupti.s@unimy.edu.my
Assessment Type	Group Project
Assessment Title	Design and Development of a Smart Community Service Management System
Total Marks	100
Weightage	20%
Submission Date	21 July 2026
CLO Mapping	• CLO2: Select a distributed computing paradigm that is suitable for a problem. • CLO3: Build distributed applications that are small in size.

1. Project Overview

This project requires students to apply distributed computing concepts, client-server architecture, database integration, and web application development techniques to design and develop a functional distributed application.

Students are expected to analyse requirements, select an appropriate distributed computing approach, design the system architecture, develop a working prototype, and demonstrate the completed solution.

2. Project Scenario

A local community centre organizes various community services and activities such as volunteer programmes, educational workshops, health campaigns, and social events.

Currently, information is managed across multiple platforms, making it difficult to track activities, registrations, and schedules.

The community centre intends to develop a **Smart Community Service Management System** that enables users to:

- View available activities
- Register for activities
- Track participation records
- View schedules and announcements

through a centralized web-based platform.

As a development team, you are required to analyse, design, develop, and demonstrate a distributed application that addresses the requirements of the proposed system.

Project Tasks

Part A: Requirement Analysis and Solution Design (20 Marks)

a) Requirement Analysis (10 Marks)

Students are required to analyse the project scenario and identify the main requirements of the proposed system.

Students should include:

1. Functional Requirements

Identify at least FIVE (5) functions the system should provide.

Examples:

- User registration and login
- View available activities
- Register for activities
- View activity schedule
- Admin manages activity records

2. Non-Functional Requirements

Identify at least THREE (3) quality requirements of the system.

Examples:

- Security
- Usability
- Performance

- Reliability
- Accessibility

Students must briefly explain why each requirement is important for the proposed system.

b) Distributed Solution Design (10 Marks)

Students are required to propose a suitable distributed solution for the system.

Students should include:

i. System Architecture

Draw and explain a simple architecture diagram showing:

- User
- Web Application
- Application Server
- Database

ii. Distributed Computing Approach

Explain the selected approach, such as:

- Client-server architecture
- Web-based distributed application
- Three-tier architecture

iii. Technology Stack

Suggest suitable technologies for development.

Examples:

- Frontend: HTML, CSS, JavaScript, Bootstrap, React
- Backend: PHP, Node.js, Java Servlet
- Database: MySQL, MariaDB, SQLite
- Tools: Visual Studio Code, XAMPP/WAMP, phpMyAdmin, Draw.io, Lucidchart, Figma

Students must justify why the selected architecture and technologies are suitable for the proposed system.

Part B: System Design and Database Design (20 Marks)

a) User Interface Design (10 Marks)

Students are required to design wireframes for at least FOUR (4) key system interfaces.

Examples:

- Login Page
- User Registration Page
- Activity Listing Page
- Activity Registration Page
- User/Admin Dashboard

Each wireframe should clearly show:

- Page title
- Main buttons
- Input fields
- Navigation menu
- Main content layout

Students may use tools such as:

- Figma
- Draw.io
- Canva
- Microsoft PowerPoint
- Lucidchart

b) Database Design (10 Marks)

Students are required to design a database structure for the proposed system.

Students should include:

- At least FOUR (4) related tables
- Suitable field names
- Appropriate data types
- Primary keys

- Foreign keys, where applicable
- An Entity Relationship Diagram (ERD)

Suggested tables:

- User
- Activity
- Registration
- Schedule

Students may use tools such as:

- MySQL Workbench
- Draw.io
- Lucidchart
- Microsoft Visio
- PowerPoint

Students must briefly explain how the tables are connected.

Part C: Application Development (50 Marks)

Develop a functional distributed web application.

Minimum Functional Requirements

User Functions

- User Registration
- User Login
- View Activities
- Register for Activities
- View Participation History
- View Activity Schedule

Administration Functions

- Manage Activities
- Manage Schedules
- View Registrations

- Generate Basic Reports
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Technical Requirements

The system should demonstrate:

- Client-Server Architecture
 - Database Connectivity
 - CRUD Functionality
 - Session Management
 - Input Validation
 - User-Friendly Interface
 - Multi-user Access
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Suggested Technologies

Students may use technologies from the list below or any suitable technologies approved by the lecturer.

Frontend

- HTML
- CSS
- JavaScript
- Bootstrap
- React

Backend

- PHP
- Node.js
- Java Servlet

Database

- MySQL
- MariaDB
- SQLite

Development Tools

- Visual Studio Code
- XAMPP/WAMP
- phpMyAdmin

Part D: Project Demonstration (10 Marks)

Each group is required to conduct a live demonstration of the developed system.

The demonstration should include:

- Brief introduction to the proposed system
- Overview of the selected system architecture
- Demonstration of key system functions
- Database interaction and data storage
- Explanation of the technologies used
- Question and Answer (Q&A) session

Students should demonstrate the successful operation of the developed prototype, including:

- User Registration
- User Login
- Activity Listing
- Registration Process
- Database Connectivity
- CRUD Functionality

Submission Requirements

Project Report (PDF)

- 2,000–2,500 words
- Include requirement analysis, system design, architecture diagram, database design (ERD), implementation details, testing results, and screenshots

Source Code

- Submit all source code and supporting files in ZIP format

Project Demonstration

- Each group must conduct a demonstration of the developed system during the scheduled project assessment session

Screenshots

Include screenshots of key system interfaces and functionalities developed in the project.

Group Contribution

- Include a summary table of each member's roles and contributions to the project.

Submission Format

- Project Report (PDF)
- Source Code and Supporting Files (ZIP)
- Group Contribution Summary table

Scoring Rubric (100 Marks)

Criteria	Excellent (80–100%)	Good (65–79%)	Satisfactory (50–64%)	Needs Improvement (0–49%)	Marks
Requirement Analysis and Solution Design	Demonstrates comprehensive analysis of requirements, appropriate identification of functional and non-functional requirements, and strong justification of the selected architecture and technology stack.	Demonstrates good analysis and justification with minor omissions.	Demonstrates basic analysis with limited justification.	Analysis is incomplete, unclear, or lacks relevance to the project.	20
System Design and Database Design	Produces clear, professional, and well-structured wireframes,	Produces appropriate designs with minor issues.	Produces basic designs with limited explanation and detail.	Designs are incomplete, inaccurate, or poorly presented.	20

Criteria	Excellent (80–100%)	Good (65–79%)	Satisfactory (50–64%)	Needs Improvement (0–49%)	Marks
	architecture diagrams, database design, and ERD with accurate explanations.				
Application Development	Fully functional distributed application that successfully meets all functional and technical requirements with minimal or no errors.	Application meets most requirements with minor issues.	Application demonstrates basic functionality but some requirements are incomplete.	Application is incomplete or fails to meet key requirements.	50
Project Demonstration	Demonstrates the system confidently and professionally, clearly explains architecture, functionality, database integration, technologies used, and effectively answers questions.	Demonstration is clear with adequate explanations and responses.	Demonstration covers ba		10
Total					100